

CASELET MAXIMA AND MINIMA_19-9-26

Directions for questions 1 to 5: These questions are based on the following information.

There are two colleges A and B in a city. The number of students in college A is 1000 more than the number of students in college B. The ratio of the number boys to girls in college A is 3 : 2 and that in college B is 4 : 1. The number of students in Arts, Commerce and Science streams in colleges A and B are in the ratio 3 : 2 : 1 and 1 : 2 : 2 respectively. In both the colleges i.e., A and B, there were students belonging to either Arts or Commerce or Science only.

If the difference between the number of students in college A who are in Arts stream and that in the Arts stream in college B is equal to 1100, find the number of students in college B.

- ☐ a) 2,000
- ☐ b) 1,500
- ☐ c) 2,600
- ☐ d) 1,800

Solution:

Let the number of students in college B be x .

Then the number of students in college A be $x + 1000$

$$\text{Given, } (x + 1000) \frac{3}{6} - x \times \frac{1}{5} = 1100$$

$$\Rightarrow \frac{x}{2} - \frac{x}{5} = 1100 \Rightarrow x = 2000.$$

Choice (A)

If the number of girls in college B is 400, then find the number of students in Science stream of college A.

- ☐ a) 600
- ☐ b) 500
- ☐ c) 1,200
- ☐ d) 1,300

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Solution:

Let the total number of students in college B be x .
Given, $\frac{1}{5}x = 400 \Rightarrow x = 2000$
Number of science students in science stream of college A
 $= \frac{1}{6}(x + 1000) = \frac{1}{6} \times 3000 = 500$.

Choice (B)

It is given that the total number of students in both the colleges together is 5,000, find the difference between the number of boys in both the colleges.

- ☐ a) 600
- ☐ b) 900
- ☐ c) 1,000
- ☐ d) 200

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Solution:

$$\begin{aligned}x + x + 1000 &= 5000 \Rightarrow x = 2000 \\ \therefore \text{Required difference} &= (x + 1000) \times \frac{3}{5} - x \times \frac{4}{5} \\ \Rightarrow \frac{1}{5}(9000 - 8000) &= \frac{1000}{5} = 200.\end{aligned}$$

Choice (D)

What is the percentage of boys in Commerce stream from college A?

- ☐ a) 20%
- ☐ b) 30%
- ☐ c) 42%
- ☐ d) Cannot be determined

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Solution:

As the break up of boys is not given in each department the answer cannot be uniquely determined.
Choice (D)

What is the total number of girls in both the college together given that the total number of boys in both the colleges together is 3400?

- ☐ a) 1,600
- ☐ b) 2,400
- ☐ c) 2,000
- ☐ d) 1,800

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Solution:

$$\frac{3}{5}(x + 1000) + \frac{4}{5}(x) = 3400$$

$$\Rightarrow \frac{7x}{5} = 2800 \Rightarrow x = 2000$$

$$\text{Total number of girls} = \frac{2}{5}(x + 1000) + \frac{1}{5}(x)$$

$$= 1600 \text{ (As } x = 2000\text{).}$$

Choice (A)

Directions for questions 6 to 10: These questions are based on the following information.

The table gives the number of employees, in four departments of a company, who have health, life, accident and critical insurance policies. The total number of people surveyed in each department is also given.

Department	Health	Life	Accident	Critical	Total
Marketing	21	27	12	8	50
Finance	28	34	19	13	42
Systems	17	22	9	6	37
Administration	23	29	17	11	56

Directions for question 6 to 10: Type in your answer in the input box provided below the question.

What is the maximum number of employees in the finance department who have all the four types of insurance policies?

Solution:

The maximum number would be the least of the four values i.e 13.
Ans : (13)

Directions for question 6 to 10: Type in your answer in the input box provided below the question.

What is the minimum number of employees in the systems department who have at least one policy?

Your Answer: Unattempted

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Solution:

As there are a total of 54 instances, and there are 22 people who have the life insurance policy, the number of employees who have at least one type of policy has to be 22.
Ans : (22)

Directions for question 6 to 10: Type in your answer in the input box provided below the question.

What is the maximum number of employees in the finance department who have any three of the four types of policies?

Your Answer: Unattempted

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Solution:

The total number of instances
= $28 + 34 + 19 + 13 = 94$.

The maximum number of employees having any three would be

$$\frac{28 + 34 + 19 + 13}{3} = \frac{94}{3} = 31, \text{ (not possible as one of the values is higher than 31).}$$

$$\frac{28 + 19 + 13}{2} = 30.$$

Ans : (30)

Directions for question 6 to 10: Type in your answer in the input box provided below the question.

What is the maximum number of employees in all the four departments together who have all the four types of policies?

Your Answer: Unattempted

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Solution:

The maximum number is $8 + 13 + 6 + 11 = 38$.

Ans : (38)

Directions for question 6 to 10: Type in your answer in the input box provided below the question.

What is the maximum number of people in the four departments who do not have any of the four types of policies?

Your Answer: Unattempted

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Solution:

The maximum number would $(50 - 27) + (42 - 34) + (37 - 22) + (56 - 29)$
 $= 23 + 8 + 15 + 27 = 73.$ Ans : (73)

Directions for questions 11 to 15: These questions are based on the following information.

In a university 1200 students are surveyed, out of which boys and girls are in the ratio 15 : 9. The survey is on the preparation for competitive exams A, B and C. Each student is preparing for one or more of the three exams. 60% of boys are preparing for A, and a total of 580 students are preparing for C. The number of boys preparing for only A is 33

$\frac{1}{3}\%$

more than those preparing for only A and B, which is 20% more than those preparing for only B and C, which is 25% more than those preparing for only C. 170 boys are preparing for both A and C. The number of girls, who are preparing for only A, only B and only C are in the ratio 6 : 8 : 7 and the number of girls who are preparing for more than one exam is 240, of which one-sixth are preparing for all the three. The number of students who are preparing for A and C is 250.

How many boys are preparing for more than one exam?

- ☐ a) 360
- ☐ b) 390
- ☐ c) 420
- ☐ d) 450

Solution:

$$\text{Total number of boys} = 1200 \times \frac{15}{24} = 750$$

$$\text{Total number of girls} = 1200 \times \frac{9}{24} = 450$$

$$n(A) = \frac{60}{100} \times 750 = 450;$$

Let Boys preparing for only A = $200x$

$\therefore$ Boys preparing for

Only A and B = $150x$

Only B and C = $125x$

Only C = $100x$

Given, 170 boys are preparing for A and C.

Among Boys:

$$n(A) = n(A \cap C) + n(\text{only A}) + n(\text{only A and B})$$

$$\therefore 450 = 170 + 200x + 150x$$

$$\Rightarrow 350x = 280 \Rightarrow x = 0.8$$

$$\therefore n(\text{only B}) = n(A \cup B \cup C) - n(A) - n(\text{only B and C}) - n(\text{only C})$$

$$= 750 - 450 - 100 - 80 = 120.$$

Among Girls:

No. of girls who are preparing for exactly one exam

$$= 450 - 240 = 210$$

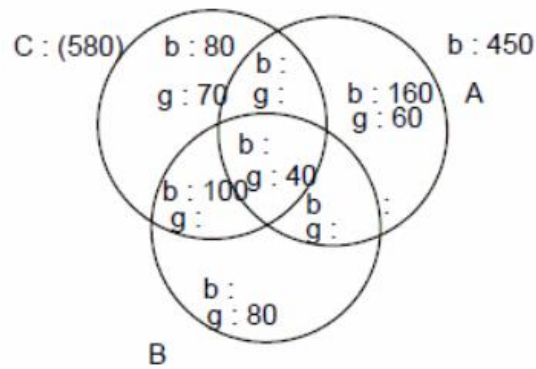
$$n(\text{only A}) : n(\text{only B}) : n(\text{only C}) = 6 : 8 : 7$$

$$\therefore n(\text{only A}) = \frac{6}{21} \times 210 = 60, n(\text{only B}) = 80, n(\text{only C}) = 70$$

$$n(A \cap B \cap C) = \frac{1}{6} \times 240 = 40$$

$\therefore$ Following Venn diagram can be drawn:

∴ Following Venn diagram can be drawn:



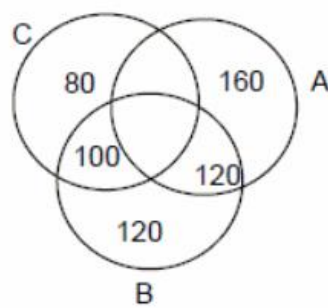
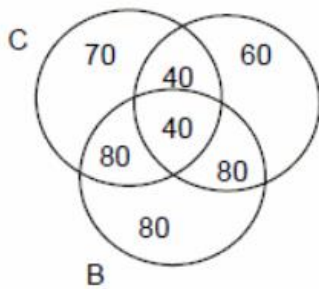
Given,

$$n(A \cap C) = 170; n(A \cap B \cap C) \text{ (for girls)} = 40$$

$$\therefore \text{For girls: } n(\text{only } A \cap C) = 250 - 170 - 70 = 40.$$

For girls:

For boys:



$$\text{Required number of boys} = n(A \cap C) \text{ only} + n(A \cap B) \text{ only} + n(B \cap C) \text{ only} \\ = 170 + 120 + 100 = 390.$$

Choice (B)

What is the difference between the number of boys and girls who are preparing for C?

☐ a) 156

☐ b) 144

☐ c) 125

☐ d) 120 (Correct Answer - ✓)

What is the ratio of the number of boys preparing for only A and that of only B?

- ☐ a) 4 : 3 (Correct Answer - ✓)
- ☐ b) 5 : 3
- ☐ c) 3 : 2
- ☐ d) 6 : 5

What percent of boys are preparing for exactly one of the three exams?

- ☐ a) 46%
- ☐ b) 48% (Correct Answer - ✓)
- ☐ c) 52%
- ☐ d) 58%

What is the number of girls preparing for only B and C?

- ☐ a) 65
- ☐ b) 70
- ☐ c) 80 (Correct Answer - ✓)
- ☐ d) 120